CAPEAI

VESSEL INVESTMENT ANALYSIS

Reference book

A complete guide to the calculator,
cash flows, assumptions, and limits.

One vessel. Explicit cash flows. Reproducible results.

Model 1.0.0 / Study schema 1 / 18 October 2026
Annual end-of-year periods / USD / Device-first storage

01 User and financial-model manual

02 Glossary and notation

03 Original system and forensic findings

04 Target system and acceptance criteria

05 Financial model contract

06 References and source boundaries

STE-inspired plain technical English. No ASD-STE100 conformity claim.

Original financial explanations. No author or institution endorsement.

Illustrative assumptions require independent commercial and tax review.

Contents

Use the page numbers or click an entry. PDF bookmarks also navigate the full reference book.

User and financial-model manual 3

1. Purpose and interpretation 3
2. Complete a study 4
3. Units, signs, and timing 6
4. Input register 7
5. Equations and model sequence 9
6. Worked examples 15
7. Saving, importing, and sharing 20
8. Local operation 23
9. Errors and unavailable results 24
10. Limits and analyst checks 24
11. Implementation reference 25
12. Editorial verification 26

Glossary and notation 27

Original system and forensic findings 33

Target system and acceptance criteria 38

Financial model contract 42

References and source boundaries 51

REFERENCE BOOK / 01

User and financial-model manual

Model version: 1.0.0.

Study schema: 1.

Date: 18 October 2026.

Scope: one vessel, one fixed-rate loan, annual cash flows, USD.

This manual describes the rebuilt calculation engine. Current system preserves the original application's behavior and defects.

The writing follows an STE-inspired editorial target. It does not claim ASD-STE100 certification or formal conformity.

1. Purpose and interpretation

CapeAI calculates the cash consequences of a vessel investment. It connects acquisition, operations, debt, fees, tax assumptions, and disposal.

The tool answers three related questions:

- What cash does the vessel generate before financing?
- What cash do equity investors contribute and receive?
- Which years require more funding under the entered assumptions?

The tool does not estimate market prices or predict charter rates. Its example contains illustrative assumptions, not current market observations.

1.1 Separate the asset from its financing

Asset cash flow measures the investment before borrowing. Equity cash flow measures the investor's remaining cash after financing payments.

Debt reduces initial equity funding. It also creates interest, principal, and maturity payments.

Higher equity IRR does not establish lower risk. A financing structure can increase both the return percentage and cash shortfalls.

Use the asset discount rate for asset cash flows. Use the equity discount rate for investor cash flows.

This separation follows the valuation principles in Damodaran's Approaches to Valuation.

1.2 Read NPV, IRR, and funding together

NPV measures discounted value after the investment cost. Positive NPV means the entered cash flows exceed the entered required return.

IRR reports a percentage return for a valid cash-flow pattern. It does not show the investment's absolute dollar value.

The equity multiple reports distributions per dollar contributed. It does not account for payment timing.

Additional equity reports modeled funding needs after acquisition. A positive lifetime profit can coexist with substantial interim funding needs.

The discount rates are assumptions. CapeAI does not derive a cost of capital or certify an acceptable investment hurdle.

Damodaran's Capital Budgeting under Certainty discusses these differences and IRR limitations.

2. Complete a study

The interface separates assumption entry, result inspection, and study management.

Navigation item	Use it to
Study setup	Enter and revise the investment assumptions.
Cash flow	Reconcile annual operations, financing, exit, and investor returns.
Scenarios	Compare charter-rate changes with the current study.
Device library	Open named studies, create a blank study, load the example, or import JSON.
Manual & equations	Open this reference and download the manual PDF.

Study setup contains Acquisition, Operations, Financing, and Exit & returns tabs.

2.1 Start and identify the investment

1. Open Device library.
2. Select New blank study or Load example.
3. Open Study setup.
4. Select Acquisition.
5. Enter a study name.
6. Enter the purchase price.
7. Set the holding period.
8. Enter acquisition transaction costs.
9. Enter initial working capital.
10. Open Vessel details & study notes.
11. Enter the vessel's identifying details.
12. Record the source and date of your assumptions in the notes.

The blank study contains no invented purchase price or operating forecast. Results remain unavailable until the required inputs are valid.

2.2 Build the forecast

1. Select Operations.
2. Enter the base daily charter rate.
3. Enter base earning days.
4. Enter daily operating costs.
5. Enter annual cost days.
6. Enter operating-cost inflation.
7. Enter brokerage and management fees.

8. Add annual charter assumptions where the base forecast differs.
9. Add each survey or capital payment to its payment year.

A zero charter rate or zero earning days produces zero revenue. An absent annual row uses the base charter assumptions.

An annual row replaces both the rate and earning-day assumptions for that year. It does not add extra revenue.

In Operations, use Add year override for an annual charter exception. Use Add capital cost for a dated payment.

If you shorten the holding period, review rows beyond the new exit year. Remove or move those rows deliberately.

The removal action names your selected holding year, such as Remove rows after Year 3.

It deletes all affected rows without another confirmation. It does not move their cash flows into another year.

2.3 Define financing and exit

1. Select Financing.
2. Enter the debt percentage of purchase price.
3. Enter the fixed annual interest rate.
4. Enter the loan tenor.
5. Enter the balloon percentage of initial debt.
6. Enter the debt arrangement fee.
7. Select Exit & returns.
8. Select the terminal-value method.
9. Enter the selected method's assumptions.
10. Enter disposal costs.
11. Enter the carry percentage.
12. Set the asset and equity discount rates.

Set debt to zero for an all-equity acquisition. There is no automatic refinancing after maturity.

2.4 Apply tax assumptions deliberately

1. Select Exit & returns.
2. Enter the modeled tax rate.
3. Enter annual tax depreciation.
4. Select whether positive disposal gains enter taxable income.
5. Check the tax assumptions against the intended investment structure.

The default tax rate is zero. This is a blank assumption, not a statement that the investment is tax-exempt.

Tax treatment requires an independent jurisdiction-specific assessment. The exact simplified formula appears in Section 5.5.

2.5 Inspect the results

1. Open Cash flow.

2. Reconcile initial debt and equity to acquisition uses.
3. Inspect each year's charter revenue.
4. Check CAPEX payment years.
5. Inspect the loan maturity payment.
6. Inspect the exit-year debt settlement.
7. Read each additional-equity warning.
8. Compare asset and investor returns.
9. Open Scenarios.
10. Inspect lower charter-rate cases.
11. Export the completed study.

Do not approve an investment from its headline IRR alone. Review its assumptions, downside cash needs, and terminal-value dependence.

3. Units, signs, and timing

Item	Convention
Currency	All monetary inputs and outputs use USD.
Daily amounts	Charter and operating rates use USD/day.
Scrap rate	USD per metric tonne of light displacement.
Percentage input	Enter 7 for 7%, not `0.07`.
Equation percentage	Divide each percentage input by 100.
Holding period	Whole years from 1 through 30.
Loan tenor	Whole years from 1 through 30.
Time zero	Acquisition funding and year-zero CAPEX.
Later cash flows	End of each model year.
Positive investor cash	A distribution to investors.
Negative investor cash	A contribution from investors.
Expense columns	Positive amounts that the cash-flow equation subtracts.
Maximum annual days	365; leap years are not modeled.
Display rounding	Presentation rounding only; the engine retains numerical precision.

Every charter row represents that future year's nominal rate. OPEX inflation does not automatically increase charter rates, management fees, CAPEX, or terminal value.

All annual amounts are netted at year end. The model cannot reveal a cash shortfall within a year.

Debt interest uses the year's opening debt balance. Principal repayments occur after that year's interest calculation.

The exit occurs after operating cash flow, scheduled principal, and any maturity payment in the final year.

Initial working capital remains committed throughout the holding period. The model releases the same amount at exit.

It does not spend this working capital to absorb interim losses. Negative annual investor cash still requires a separate equity contribution.

4. Input register

The field paths below also identify values in the exported study JSON. Technical bounds prevent unsupported calculations; they do not establish commercial plausibility.

Text must not contain unsupported control characters or incomplete Unicode characters. Normal tabs, line breaks, and complete Unicode characters are permitted.

4.1 Identification and acquisition

Field	Meaning	Unit or limit
`schemaVersion`	Study data structure	Must equal 1.
`name`	Study name	Text, maximum 200 characters.
`notes`	Assumption sources and analyst notes	Text, maximum 10,000 characters.
`vessel.name`	Vessel name	Text, maximum 200 characters.
`vessel.type`	Vessel type	Text, maximum 200 characters.
`vessel.builtYear`	Descriptive build year	Whole number, 0-2200; 0 can mean unspecified.
`vessel.deadweight`	Descriptive deadweight tonnage	Tonnes, 0-1,000,000.
`vessel.ldt`	Light displacement for scrap calculation	Tonnes, 0-1,000,000.
`acquisition.price`	Vessel purchase price	USD, 0.01-1,000,000,000,000.
`acquisition.transactionCosts`	Initial transaction costs	USD, 0-1,000,000,000,000.
`acquisition.workingCapital`	Initial funding released at exit	USD, 0-1,000,000,000,000.

Name, type, build year, and deadweight do not change returns. LDT changes returns only when scrap mode sets the terminal value.

4.2 Financing and projection

Field	Meaning	Unit or limit
`financing.ltvPct`	Initial debt as a percentage of purchase price	0-100%.
`financing.interestPct`	Fixed annual interest on opening debt	0-100%.
`financing.tenorYears`	Year of full loan maturity	Whole years, 1-30.
`financing.balloonPct`	Initial debt fraction excluded from regular amortization	0-100%.
`financing.arrangementFeePct`	Fee on initial debt draw	0-100%.
`projection.holdYears`	Year of vessel disposal	Whole years, 1-30.
`projection.assetDiscountPct`	Required return for asset NPV	0-100% annually.
`projection.equityDiscountPct`	Required return for equity NPVs	0-100% annually.

Debt funds only its specified percentage of purchase price. Transaction costs, working capital, CAPEX, and loan fees increase the remaining equity requirement.

4.3 Operations and schedules

Field	Meaning	Unit or limit
`operations.dailyOpex`	First-year operating cost per cost day	USD/day, 0-100,000,000.
`operations.opexDays`	Annual days that accrue operating cost	Whole days, 0-365.
`operations.inflationPct`	Annual OPEX change after year one	-100-100%.
`operations.baseDailyRate`	Charter rate when no annual override exists	USD/day, 0-100,000,000.
`operations.baseRevenueDays`	Earning days when no annual override exists	Whole days, 0-365.
`operations.brokeragePct`	Percentage of gross charter revenue	0-100%.
`operations.managementFeePct`	Annual percentage of original purchase price	0-100%.
`charter[].year`	Year to override	Whole year, 1 through holding period.
`charter[].dailyRate`	Override daily charter rate	USD/day, 0-100,000,000.
`charter[].revenueDays`	Override earning days	Whole days, 0-365.
`capex[].year`	Year of capital payment	Whole year, 0 through holding period.
`capex[].amount`	Capital payment	USD, 0-1,000,000,000,000.
`capex[].label`	Payment description	Text, maximum 200 characters.

Each charter year can appear once. Several CAPEX payments may occur in the same year; the engine adds them.

The engine and native importer accept at most 30 charter rows and 300 CAPEX rows.

OPEX and earning days are independent assumptions. Reduced earning days do not automatically reduce operating costs.

Avoid double counting technical management costs. Enter them in daily OPEX or the separate management fee according to their actual contractual basis.

4.4 Exit, tax, and carry

Field	Meaning	Unit or limit
`exit.mode`	Terminal-value method	`depreciation`, `manual`, or `scrap`.
`exit.depreciationPct`	Annual decline used only by declining-value exit	0-100%.
`exit.manualValue`	Gross vessel sale value for manual exit	USD, 0-1,000,000,000,000.
`exit.scrapPerTon`	Scrap price applied to LDT	USD/tonne, 0-100,000,000.
`exit.disposalCostPct`	Selling costs on gross terminal value	0-100%.
`tax.ratePct`	Simplified annual tax rate	0-100%.
`tax.annualDepreciationPct`	Annual declining-balance tax depreciation	0-100%.
`tax.taxSaleGain`	Include positive sale gains in final-year taxable income	`true` or `false`.
`carry.ratePct`	Sponsor share of positive aggregate nominal equity profit	0-100%.

Exit-value decline and tax depreciation are separate inputs. Neither is an appraisal or an automatic tax-law assumption.

Inactive exit fields remain in the study. Only the selected exit mode determines gross terminal value.

5. Equations and model sequence

5.1 Symbols

All currency variables below use USD unless another unit is stated. Rates in equations are decimal fractions.

Symbol	Definition
t, H, T	Current year, holding years, loan tenor years.
P, X, W	Purchase price, transaction costs, initial working capital.
K_t	Sum of CAPEX paid at time t.
λ, i, b, a	Debt fraction, interest rate, balloon fraction, arrangement-fee rate.
D_0, E_0	Initial debt and equity funding.
F_D	Initial debt arrangement fee.
h_t, e_t	Charter rate in USD/day and earning days in year t.

Symbol	Definition
o, d_o, g	First-year daily OPEX, annual cost days, OPEX inflation.
β, m, c	Brokerage rate, annual management-fee rate, carry rate.
R_t, O_t, B_t, M_t	Revenue, OPEX, brokerage, management fee.
S_t	Operating surplus after brokerage and management.
L_t, I_t	Opening debt and annual interest.
A, A_t	Scheduled annual principal amount and actual scheduled payment.
J_t, Q_t	Maturity principal and additional exit debt settlement.
V, s, N	Gross terminal value, disposal-cost rate, net sale cash.
δ, q, ℓ	Exit-value decline rate, scrap price per tonne, vessel LDT.
τ, p, TB_t, Dep_t	Tax rate, tax-depreciation rate, closing tax book, tax depreciation.
G_t	Positive taxable sale gain, if enabled.
TaxA_t, TaxE_t, TaxF_t	Asset tax, equity tax, and pre-annual-fee equity tax.
Z_t	Net sale cash plus working-capital release.
ACF_t, FCF_t, PCF_t, ECF_t	Asset, pre-annual-fee equity, pre-carry equity, and final investor cash flow.
C	Total carry paid in the final year.
r_A, r_E	Asset and equity discount rates.

5.2 Acquisition funding

```
K_0 = sum of all CAPEX entries assigned to year 0

D_0 = P × λ

F_D = D_0 × a

Asset uses = P + X + W + K_0

Total initial uses = P + X + W + K_0 + F_D

E_0 = Total initial uses - D_0
```

The initial funding identity is:

```
Initial debt + Initial equity = Total initial uses
```

The asset stream excludes the financing arrangement fee. All equity streams include that fee in initial equity.

5.3 Annual revenue, costs, and fees

For an overridden year, use its entered `h_t` and `e_t`. Otherwise, use the base rate and base earning days.

```
R_t = h_t × e_t

O_t = o × d_o × (1 + g)^(t - 1)

B_t = R_t × β

M_t = P × m

S_t = R_t - O_t - B_t - M_t

K_t = sum of CAPEX entries assigned to year t
```

Year one has no OPEX inflation uplift. Year two applies one uplift; year three applies two.

Management fees use original purchase price every year. They do not decline with debt or terminal value.

CAPEX is a cash payment when scheduled. The model does not spread that payment across the holding period.

5.4 Debt and terminal value

The regular principal schedule is equal-principal amortization, not an annuity payment.

```
A = D_0 × (1 - b) / T

L_1 = D_0

I_t = L_t × i

A_t = min(L_t, A), when t ≤ T; otherwise 0

Debt after scheduled principal = max(0, L_t - A_t)

J_t = Debt after scheduled principal, when t = T; otherwise 0

Debt after maturity payment = max(0, L_t - A_t - J_t)

Q_t = Debt after maturity payment, when t = H; otherwise 0

Closing debt = max(0, L_t - A_t - J_t - Q_t)

L_(t+1) = Closing debt

Total principal_t = A_t + J_t + Q_t

Total debt service_t = I_t + A_t + J_t + Q_t
```

The sale payment `Q_t` is additional to scheduled principal. If maturity occurs at exit, `J_t` already clears the balance.

If maturity precedes exit, later debt and interest are zero. The model does not insert a replacement loan.

Gross terminal value follows the selected mode:

```
Declining-value mode: V = P × (1 - δ)^H

Manual mode: V = entered manual value

Scrap mode: V = ℓ × q

Disposal costs = V × s

N = V - Disposal costs
```

Terminal value is a sale assumption, not a perpetuity. Transaction costs and later CAPEX do not automatically increase it.

Net sale cash is calculated before debt settlement and tax. The annual cash-flow equations deduct those items separately.

5.5 Simplified tax

The initial tax book includes purchase price, transaction costs, and initial CAPEX. It excludes working capital and the loan arrangement fee.

```
TB_0 = P + X + K_0

Dep_t = TB_(t-1) × p

TB_t = max(0, TB_(t-1) - Dep_t) + K_t

G_H = max(0, N - TB_H), when sale-gain taxation is enabled

G_t = 0 in every other case

TaxA_t = τ × max(0, S_t - Dep_t + G_t)

TaxE_t = τ × max(0, S_t - I_t - Dep_t + G_t)

TaxF_t = τ × max(0, R_t - O_t - I_t - Dep_t + G_t)
```

Annual CAPEX enters the tax book after that year's depreciation. It begins depreciating in the following year.

Asset tax excludes the interest deduction. Equity tax includes it.

The pre-annual-fee stream removes brokerage and management fees. Its tax must therefore be recalculated on the higher taxable operating result.

A positive sale gain enters final-year taxable income. A sale below tax book does not create a modeled deductible sale loss.

Within the final year, a negative operating tax base can offset a positive modeled sale gain. The final tax amount cannot be negative.

There are no refunds, loss carryforwards, tax credits, tonnage taxes, VAT, withholding taxes, or investor-level taxes.

Debt arrangement costs receive no tax deduction in this model. Principal repayment and working-capital release do not enter taxable income.

This is an explicit screening convention. It is not a jurisdiction-specific tax calculation.

5.6 The four cash-flow streams

At time zero:

```
ACF_0 = -(P + X + W + K_0)

FCF_0 = PCF_0 = ECF_0 = -E_0
```

During operation:

```
Z_t = 0, for t < H

Z_H = N + W

ACF_t = S_t - K_t - TaxA_t + Z_t

FCF_t = R_t - O_t - K_t - TaxF_t - I_t - A_t - J_t - Q_t + Z_t

PCF_t = S_t - K_t - TaxE_t - I_t - A_t - J_t - Q_t + Z_t
```

Stream	Annual fees	Debt	Carry	NPV rate
Asset, ACF	Included	Excluded	Excluded	Asset rate.
Pre-annual-fee equity, FCF	Excluded	Included	Excluded	Equity rate.
Pre-carry equity, PCF	Included	Included	Excluded	Equity rate.
Final investor, ECF	Included	Included	Included	Equity rate.

“Pre-fees” in engine output means before annual brokerage and management fees. It still includes transaction costs, disposal costs, and debt arrangement fees.

The asset stream includes the same operating management and brokerage charges as the vessel investment. It excludes investor carry and all financing effects.

5.7 Carry

```
Aggregate pre-carry profit = Σ PCF_t, for t = 0 through H

C = c × max(0, Aggregate pre-carry profit)

ECF_t = PCF_t, for t < H

ECF_H = PCF_H - C
```

The initial investment and later negative cash flows reduce the carry base. Carry is zero when aggregate pre-carry profit is zero or negative.

Carry has no preferred-return hurdle, catch-up tier, investor split, escrow, or clawback model.

Carry is charged at exit even if earlier distributions generated the aggregate profit. This can increase the final year's equity contribution.

5.8 Returns and funding

For any cash-flow stream `CF_t`, CapeAI calculates:

```
Invested capital = -Σ min(CF_t, 0), for t = 0 through H

Distributions = Σ max(CF_t, 0), for t = 0 through H

Nominal profit = Distributions - Invested capital

Multiple = Distributions / Invested capital

ROI = Nominal profit / Invested capital

NPV(r) = Σ [CF_t / (1 + r)^t], for t = 0 through H
```

Multiple and ROI are unavailable when invested capital is zero. ROI is a total holding-period percentage, not an annual return.

Asset NPV uses `r_A`. All three equity NPVs use `r_E`.

IRR solves this equation:

```
0 = Σ [CF_t / (1 + IRR)^t], for t = 0 through H
```

The engine requires finite cash flows, a negative investment, and a positive distribution. It ignores zero cash flows when counting sign changes.

More than one sign change produces an ambiguous status. This conservative rule does not assert that multiple roots always exist.

A positive first nonzero cash flow is outside the supported investment-IRR convention. A missing valid numerical solution produces an unavailable status.

The solver brackets and bisects `log(1 + IRR)`. It verifies a scaled residual before returning a rate above -100%.

Annual investor funding is calculated as follows:

```
Contribution_t = max(0, -ECF_t), for t ≥ 1

Distribution_t = max(0, ECF_t), for t ≥ 1

Additional equity = Σ Contribution_t, for t = 1 through H

Cumulative investor cash_t = Σ ECF_j, for j = 0 through t

Total investor contributions = E_0 + Additional equity
```

These are gross contributions and distributions under immediate annual distribution. Earlier positive cash is not retained to offset a later deficit.

5.9 Debt-service coverage

The coverage calculation excludes vessel sale proceeds and working-capital release. It also excludes sale-gain tax and carry.

```
Operating tax_t = τ × max(0, S_t - I_t - Dep_t)

CFADS_t = S_t - K_t - Operating tax_t

Contractual debt service_t = I_t + A_t + J_t

DSCR_t = CFADS_t / Contractual debt service_t
```

The denominator includes the maturity balloon. It excludes debt repaid only because of early disposal.

The numerator deducts actual scheduled CAPEX. It uses the operating equity-tax assumption, which includes an interest deduction.

DSCR is unavailable when contractual debt service is zero. It is not displayed as infinity.

A ratio below 1 means operating cash does not cover the defined contractual payments. Disposal cash may still fund the final-year payment.

A lender's covenant definition can differ. Use the facility agreement for covenant compliance.

5.10 Daily charter breakeven

The breakeven rate covers OPEX, annual fees, CAPEX, operating tax, interest, and contractual principal.

It excludes early-sale debt settlement, sale proceeds, working-capital release, sale-gain tax, and carry.

Define the following values for each year:

```
Cash need = K_t + I_t + A_t + J_t

Tax shield = I_t + Dep_t

Net revenue share = 1 - β
```

The required operating surplus is piecewise:

```
If Cash need ≤ Tax shield:
    Required surplus = Cash need
If Cash need > Tax shield and τ < 1:
    Required surplus = (Cash need - τ × Tax shield) / (1 - τ)
Breakeven daily rate = (O_t + M_t + Required surplus) / [e_t × (1 - β)]
```

The result is unavailable with zero earning days or 100% brokerage. It is also unavailable when 100% tax prevents finite coverage.

This is a one-year cash-coverage rate. It does not recover the initial investment or achieve the equity discount rate.

5.11 Scenarios and totals

The charter sensitivity uses multipliers of 0.70, 0.85, 1.00, 1.15, and 1.30.

```
Scenario base rate = Original base rate × multiplier
Scenario annual rate_t = Original annual rate_t × multiplier
```

Earning days, OPEX, debt terms, and other assumptions remain unchanged in this comparison. Each case runs through the full engine.

Changing revenue therefore changes brokerage, taxable income, carry, funding needs, and returns where applicable.

The scenarios have no assigned probabilities. A 30% reduction is a selected stress, not an estimated confidence interval.

Totals sum their corresponding annual columns. Total CAPEX also includes year-zero CAPEX.

Total principal must equal the original debt draw. Closing debt must be zero after the final model year.

6. Worked examples

All examples use invented inputs. Outputs below were calculated with model version 1.0.0.

6.1 All-equity investment: profit with negative NPV

Use the following inputs. Set all unlisted costs, fees, tax, inflation, debt, and carry to zero.

Input	Value
Purchase price	$10,000,000
Holding period	2 years
Daily charter rate	$10,000
Earning days	350
Daily OPEX	$6,000
Cost days	365
Manual sale value	$9,000,000
Asset and equity discount rates	10%

Annual revenue is `$10,000 × 350 = $3,500,000`. Annual OPEX is `$6,000 × 365 = $2,190,000`.

Annual operating cash is $1,310,000. The year-two sale adds $9,000,000.

Time	Investor cash flow
0	-$10,000,000
1	$1,310,000
2	$10,310,000

```
NPV = -10,000,000 + 1,310,000/1.10 + 10,310,000/1.10²
    = -$288,429.75
IRR = 8.2992%
Equity multiple = 11,620,000 / 10,000,000 = 1.1620×
Nominal profit = $1,620,000
Breakeven hire = 2,190,000 / 350 = $6,257.14/day
```

The study has positive nominal profit. It fails to earn the entered 10% required return.

Asset and equity returns match because the study has no financing or carry.

6.2 Early sale: repay the actual loan balance

Start with Example 6.1. Change the holding period to three years and manual sale value to $8 million.

Set debt to 50%, interest to 6%, tenor to five years, and balloon to 40%.

Retain Example 6.1's operating assumptions. Set additional costs, fees, tax, inflation, and carry to zero.

```
Initial debt = $10,000,000 × 50% = $5,000,000
Initial equity = $5,000,000
Annual scheduled principal = $5,000,000 × (1 - 40%) / 5 = $600,000
```

Item	Year 1	Year 2	Year 3
Opening debt	$5,000,000	$4,400,000	$3,800,000
Interest	$300,000	$264,000	$228,000
Scheduled principal	$600,000	$600,000	$600,000
Additional sale settlement	$0	$0	$3,200,000
Closing debt	$4,400,000	$3,800,000	$0
Investor cash	$410,000	$446,000	$5,282,000

The final-year investor cash is:

```
1,310,000 - 228,000 - 600,000 - 3,200,000 + 8,000,000
= $5,282,000
```

The original balloon assumption is $2 million. Early sale requires $3.2 million after the scheduled payment.

Investor IRR is 7.6483%. Investor NPV at 10% is -$290,232.91. The equity multiple is 1.2276×.

Year-three DSCR is `1,310,000 / 828,000 = 1.5821×`. It excludes both the sale proceeds and early-sale debt settlement.

6.3 Tax, fees, CAPEX, working capital, and carry

Use a one-year holding period with these inputs:

Input	Value
Purchase price	$1,000,000
Transaction costs	$20,000
Initial working capital	$50,000
Year-zero CAPEX	$30,000
Year-one CAPEX	$40,000
Debt percentage	50%
Interest / tenor / balloon	8% / 5 years / 20%
Debt arrangement fee	1%
Charter rate / earning days	$2,000 / 300
Daily OPEX / cost days	$500 / 360
Brokerage / management	5% / 2%
Manual sale value / disposal cost	$1,100,000 / 2%
Tax rate / tax depreciation	20% / 10%
Sale-gain taxation	Enabled
Carry	20%
Asset / equity discount rates	10% / 12%

Set inflation to zero. Acquisition funding is:

```
Debt = $500,000
Arrangement fee = $5,000
Asset uses = 1,000,000 + 20,000 + 50,000 + 30,000 = $1,100,000
Initial equity = 1,100,000 + 5,000 - 500,000 = $605,000
```

Year-one operations and tax are:

```
Revenue = 2,000 × 300 = $600,000
OPEX = 500 × 360 = $180,000
Brokerage = 600,000 × 5% = $30,000
Management = 1,000,000 × 2% = $20,000
Operating surplus = 600,000 - 180,000 - 30,000 - 20,000 = $370,000
Interest = 500,000 × 8% = $40,000
Tax depreciation = 1,050,000 × 10% = $105,000
Closing tax book = 1,050,000 - 105,000 + 40,000 = $985,000
Net sale cash = 1,100,000 × 98% = $1,078,000
Positive sale gain = 1,078,000 - 985,000 = $93,000
Equity tax = (370,000 - 40,000 - 105,000 + 93,000) × 20% = $63,600
Asset tax = (370,000 - 105,000 + 93,000) × 20% = $71,600
```

Scheduled principal is $80,000. Early sale settles another $420,000, so total principal is $500,000.

```
Pre-carry equity cash = 370,000 - 40,000 - 63,600 - 40,000 - 500,000 + 1,078,000 + 50,000
                      = $854,400
Aggregate pre-carry profit = 854,400 - 605,000 = $249,400
Carry = 249,400 × 20% = $49,880
Final investor cash = 854,400 - 49,880 = $804,520
Investor IRR = 804,520 / 605,000 - 1 = 32.9785%
Investor NPV at 12% = -605,000 + 804,520 / 1.12 = $113,321.43
```

The equation contains two $40,000 deductions. One is CAPEX; the other is interest.

The asset stream is `[-1,100,000, 1,386,400]`. Its IRR is 26.0364%, with NPV of $160,363.64 at 10%.

Operating tax excludes the disposal gain: `(370,000 - 40,000 - 105,000) × 20% = $45,000`.

CFADS is `370,000 - 40,000 - 45,000 = $285,000`. DSCR is `285,000 / (40,000 + 80,000) = 2.3750×`.

6.4 The editable Aegean example

The application example has a $16 million purchase price and $8 million initial debt. Its holding period and loan tenor are five years.

It includes $320,000 transaction costs, $500,000 working capital, and an $80,000 loan arrangement fee. Initial equity is $8.9 million.

Annual charter rates and earning days vary. Year three includes a $1 million survey payment.

Output	Expected value
Asset IRR	16.7741%
Asset NPV at 8%	$5,452,348.62
Investor IRR after carry	19.7261%
Investor NPV at 12%	$2,455,196.42
Investor multiple	1.977831×
Gross terminal value	$12,380,495.00
Disposal costs	$247,609.90
Total carry	$2,175,674.14
Additional equity	$0.00

The fifth-year scheduled principal is $1.12 million. The maturity payment is another $2.4 million.

Year-five operating DSCR is about 1.0058×. This near-one ratio deserves attention despite the positive lifetime returns.

The example's tax rate is zero. Its result does not establish a tax treatment for a real investment.

6.5 When IRR is unavailable or ambiguous

For cash flows `[-100, 0]`, the investor recovers nothing. The engine reports IRR as unavailable.

For `[-100, 230, -132]`, cash flows change sign twice. The mathematical equation has roots at 10% and 20%.

The engine reports an ambiguous status instead of selecting one rate. Use NPV at a justified discount rate.

6.6 Additional equity reduces the carry base

Use a $100 purchase, a two-year hold, $20 year-one CAPEX, and a $170 manual sale value.

Set revenue, operating costs, debt, fees, and tax to zero. Set carry to 20%.

```
Pre-carry cash flows = [-100, -20, 170]
Whole-deal pre-carry profit = -100 - 20 + 170 = $50
Carry = $50 × 20% = $10
Investor cash flows = [-100, -20, 160]
Total contributions = $100 + $20 = $120
Additional equity = $20
Distributions = $160
Investor multiple = $160 / $120 = 1.3333×
```

The $20 contribution is part of invested capital. Ignoring it would overstate both the carry base and the investor multiple.

6.7 Maturity before disposal requires funding

Use a $100 purchase, 50% debt, two-year loan tenor, 40% balloon, and three-year holding period.

Set interest, costs, revenue, fees, tax, and carry to zero. Set the manual sale value to $100.

```
Initial debt = $50

Initial equity = $50

Scheduled principal = $50 × (1 - 40%) / 2 = $15

Year-one principal = $15

Year-two principal = $15 scheduled + $20 maturity = $35

Year-three debt = $0

Investor cash flows = [-50, -15, -35, 100]

Additional equity = $50

Investor IRR = 0%
```

The sale occurs after the loan matures. Investors must fund the intervening principal payments under this model.

The calculation does not silently postpone the balloon until sale.

7. Saving, importing, and sharing

7.1 Device-first data model

This release has no account and no cloud study database. It has no external AI integration or market-data connector.

The application saves study data in the current browser's local storage. A named library can hold up to 40 studies.

Storage belongs to the website origin, browser profile, and device. Another browser or device will not automatically see the same studies.

An origin is its scheme, host, and port; URL paths do not create storage boundaries.

GitHub Pages apps under the same `username.github.io` share an origin, so their scripts can access the same browser storage.

For confidential studies, use a separate browser profile or a trusted local server.

Browser storage is not encrypted by CapeAI. Someone with access to the same unlocked browser profile may access its studies.

Clearing site data can remove the local draft and library. Private browsing, browser policies, or storage limits can prevent saving.

The hosting service still receives ordinary requests for the application files. Study fields are not submitted to a cloud study service.

7.2 Save the draft and a named study

The editable draft saves automatically after changes when browser storage is available. A named library entry is a separate saved snapshot.

1. Enter a recognizable study name.
2. Select Save study.
3. Open Device library.
4. Check that the named entry appears.

Use Save a copy to create another named entry. This supports separate base, downside, and alternative-exit studies.

Replacing unsaved work can open Keep a copy of this study? Choose Save & continue to preserve a named copy first.

Continue without a copy proceeds without making that copy. Cancel returns to the current study.

If saving fails, the dialog shows the error and keeps the current study unchanged.

Automatic draft saving is not a portable backup. Export JSON before changing browsers, clearing site data, or moving to another device.

7.3 Keep a portable backup

1. Open Export.
2. Select Study JSON.
3. Save the file in your chosen folder.
4. Include a study date in the filename.
5. Keep the file with its supporting assumptions.

JSON contains readable study data. It is not password-protected or encrypted.

The native export identifies `capeai.study`, export version 1, and study schema version 1. It includes the export timestamp and supported study fields.

7.4 Import a study

1. Open Device library.
2. Select Import JSON.
3. Choose the exported study file.
4. Read the import warnings.
5. Correct any invalid inputs.
6. Review the calculated results.
7. Select Save study.

The import limit is 2,097,152 bytes, or 2 MiB. Unsupported native study fields are excluded with a warning.

An unsupported schema version or wrong data type blocks import. Supported data with invalid numerical bounds can load for correction.

Results remain unavailable until all blocking model errors are fixed. Import validation does not authenticate a file's author or assumptions.

7.5 Migrate original HTML studies

The importer recognizes original flat-field JSON exports. It maps supported acquisition, financing, cost, charter, and exit fields.

Original authentication fields and unsupported content are excluded. They do not enter the replacement study schema.

Legacy survey budgets lack reliable payment years. The migration assigns recognized budgets to year one and displays a warning.

1. Open the migrated CAPEX schedule.
2. Set each payment's actual year.

3. Check all annual charter rows.
4. Review the new transaction-cost and working-capital fields.
5. Review discount rates, taxes, and disposal costs.
6. Export the corrected study in the new format.

New fields use the blank-study defaults during migration. These defaults require review.

Empty legacy exit depreciation, OPEX inflation, and brokerage fields use 0%, 2.5%, and 1.25%, respectively. Explicit zero values remain zero.

The migration reports these assumptions. Review the values before comparing the imported study with an earlier report.

The old React application's separate JSON structure is not the original flat-field format. Unsupported files must be entered manually.

The importer does not remove credentials from old files or historical cloud records. Keep the historical security finding separate from the new app's behavior.

7.6 Explain the shared result

A study file preserves inputs. A report records an interpreted result at a particular model version.

Include both when another analyst must reproduce the work. Include the underlying evidence for rates, costs, financing, and terminal value.

Use the application's print view for a browser-generated PDF. PDF appearance can vary with browser print settings.

1. Open Export.
2. Select Print / PDF report.
3. Open the browser's Print command in the report tab.
4. Select the browser's PDF destination.
5. Save the report with the study JSON.

Download manual PDF opens the reference book. Print / PDF report creates a report for the current study.

Output	Contents and purpose
Study JSON	Supported editable inputs, format version, and export timestamp. Use it to reopen or reproduce a study.
Cash-flow CSV	Time-zero and annual rows, four return bases, full input paths, and model warnings. Use it for reconciliation.
Print report	Formatted returns, funding, cash flows, scenarios, and complete assumptions. Use the browser's Print command for PDF.
PowerPoint deck	Presentation summary of returns, funding, annual cash flows, scenarios, and core assumptions. Keep JSON and CSV beside it.

CSV IRR and ROI values use decimal fractions. A value of `0.10` means 10%; input percentage fields still use whole percentages.

The CSV's year-zero cash-flow columns include the complete initial funding. Its expense columns describe operating years and show zero at acquisition.

Initial CAPEX remains in the input record and initial cash outflow. Do not infer total CAPEX from the annual expense column alone.

CSV text that could start a spreadsheet formula receives a protective apostrophe. Numeric amounts remain numeric.

Reports and presentation decks recalculate the selected study with the same engine. They do not use the original application's separate export formulas.

PowerPoint creation runs locally with a bundled library. It does not send the study to a presentation service.

Exported calculated results require valid model inputs. JSON can preserve supported draft data that still needs input corrections.

The original sign-in, cloud saving, and unfinished AI actions have been retired. They are not part of this release.

8. Local operation

The application is static browser software. It does not require a study server, database, or API key.

Use Node.js 20 or later. Node.js 24 is recommended for the documented workflow.

No dependency installation is required to run the application. The repository contains its browser assets and development scripts.

The source repository is Cavramoudis/capeai-maritime.

The hosted application is CapeAI. Its web manual and PDF reference book accompany the calculator.

1. Clone the published repository.
2. Open its directory in a terminal.
3. Run the documented local server command.
4. Open the displayed local URL.
5. Import your exported study JSON.

For a new local copy, run:

```
git clone https://github.com/Cavramoudis/capeai-maritime.git
cd capeai-maritime
npm run dev
```

Open `http://127.0.0.1:4173` in your browser. Keep the terminal running while you use the application.

Press `Ctrl+C` in the terminal to stop the server. For later sessions, run `npm run dev` from the existing repository directory.

Use the same local address and port to reopen existing local studies. A different host or port creates a separate storage origin.

To run the automated checks:

```
npm test
```

To create the static deployment output:

```
npm run build
```

Local and hosted URLs have separate browser storage. Use JSON export and import to move studies between them.

The local server serves only this computer through `127.0.0.1`. It is not configured to share the calculator with other devices.

After source edits, rebuild with `npm run build` and reload the browser. The production files are generated in `dist/`.

9. Errors and unavailable results

Message or state	Meaning	Action
Invalid purchase price	The model needs a positive price.	Enter the actual study price.
Invalid number	A value is missing, nonfinite, or outside supported bounds.	Correct the identified field.
Duplicate charter year	Two rows describe the same year.	Keep one row for that year.
Schedule beyond holding period	A charter or CAPEX row occurs after disposal.	Correct the year or holding period.
IRR unavailable	The cash-flow pattern or numerical result does not support an investment IRR.	Read the detailed reason and inspect NPV.
IRR ambiguous	Cash flows change sign more than once.	Use NPV and inspect funding dates.
DSCR unavailable	Contractual debt service is zero.	Inspect operating cash directly.
Breakeven unavailable	Earning days or net revenue share prevent a finite rate.	Inspect the year's assumptions.
Additional equity required	One or more annual investor cash flows are negative.	Arrange funding or revise assumptions.
Storage unavailable	The browser cannot retain the study.	Export a JSON backup.
Damaged saved data	Some records cannot load safely; the original browser data remains unchanged.	Export readable studies before changing site storage.
Unsupported import version	The file uses an unknown schema or export version.	Use a supported version or enter the study manually.

A blank cell or unavailable result is not zero. Zero is a valid numerical result when the model calculates it.

10. Limits and analyst checks

10.1 Model boundaries

- Annual timing can conceal monthly and seasonal cash shortages.
- The model uses one vessel and one fixed-rate loan.
- It does not simulate refinancing, floating interest, hedges, or foreign exchange.
- It does not model lender covenants, enforcement, defaults, or limited-recourse loss allocation.
- It does not model a debt-service reserve account or cash sweep.
- Working capital is a fixed initial amount with full terminal recovery.
- It does not forecast receivables, payables, inventory, or working-capital losses.
- All capital payments enter one declining tax book under the simplified tax convention.

- It does not model component-specific tax lives or special survey amortization rules.
- It does not apply a preferred-return waterfall or investor-specific tax treatment.
- It does not include emissions, fuel, or regulatory costs unless the analyst enters their cash effect.
- It does not fetch market data or assign scenario probabilities.
- It assumes full collection of entered charter revenue and full recovery of entered sale proceeds.

Negative investor cash represents another contribution. It does not automatically represent a legal obligation or feasible funding commitment.

The annual model also nets inflows against outflows within the same year. Actual payment dates can require more peak liquidity than the displayed contribution.

10.2 Validate the investment case

1. Confirm the vessel's technical condition.
2. Verify the acquisition price and transaction costs.
3. Check charter counterparty and collection assumptions.
4. Estimate off-hire consistently with maintenance plans.
5. Verify operating-cost coverage and fee definitions.
6. Check the loan's actual repayment terms.
7. Confirm the tax structure independently.
8. Test lower charter rates and fewer earning days.
9. Test higher costs and earlier CAPEX.
10. Test a lower sale value.
11. Confirm the source of every required equity contribution.
12. Record the model version with the investment decision.

These checks concern the assumptions supporting the calculation. Software correctness alone cannot establish their reliability.

11. Implementation reference

File	Responsibility
src/model.mjs	Validation, annual calculations, return metrics, and sensitivity.
src/studies.mjs	Import, export, legacy migration, and local storage.
docs/MODEL-CONTRACT.md	Concise engineering contract for the model.
docs/CURRENT-SYSTEM.md	Evidence-based description of the original application.
docs/TARGET-SYSTEM.md	Product requirements and acceptance criteria.
docs/GLOSSARY.md	Controlled project terminology.

`calculateStudy()` returns validation issues before calculating an invalid study. A valid result contains funding, annual rows, four cash-flow streams, metrics, totals, and warnings.

`calculateSensitivity()` calls the same financial engine for each charter case. Reports must use these results instead of reimplementing equations.

Study schema version and model version serve different purposes. The schema identifies data structure; the model version identifies calculation rules.

The manual's equations are original explanations of this implementation. The financial references provide general valuation principles, not an endorsement of the software.

12. Editorial verification

The project's target is at least 80% of prose sentences with no more than 25 words.

At release, 100% of measured prose sentence candidates met the 25-word limit. Run `npm run check:docs` for current counts.

All measured procedural sentences contained no more than 20 words. The procedures use one principal action per step.

The repository check excludes headings, tables, fenced equations, and code samples. A separate editorial check included punctuated table cells and numbered procedures.

The check normalized Markdown links and retained inline field names. It counted words with punctuation-based sentence boundaries.

This verifies a bounded writing metric. It does not certify the ASD-STE100 grammar rules, dictionary, or full standard.

Read the writing-source note for the distinction between this editorial target and formal conformity.

REFERENCE BOOK / 02

Glossary and notation

Use these definitions when reading the application and manual. Some financial terms have different definitions in other products or loan agreements.

The model manual gives the exact equations and calculation boundaries.

Term	Meaning in CapeAI
Acquisition	The purchase of the vessel and the associated initial investment.
Acquisition costs	Transaction costs entered separately from the vessel purchase price.
Additional equity	New investor cash required when an annual investor cash flow is negative.
Annual period	One model year. Operating cash flows occur at its end for return calculations.
Asset cash flow	Cash generated by the vessel investment before debt financing and investor carry. It includes operating fees and unlevered tax assumptions.
Asset discount rate	The required annual return used to discount asset cash flows.
Asset IRR	The internal rate of return calculated from the asset cash-flow stream.
Asset NPV	The net present value calculated from asset cash flows and the asset discount rate.
Assumption	A user-supplied estimate or contractual input. The application does not verify its commercial accuracy.
Balloon	The initial debt fraction left outside scheduled equal-principal amortization. It becomes payable at maturity or earlier sale.
Base case	The study's unchanged assumptions used as the comparison reference.
Basis	The specific cash-flow stream, cost definition, or amount used in a calculation.
Book value	The remaining modeled tax basis after tax depreciation and CAPEX additions. It is not a vessel market valuation.
Breakeven charter rate	The daily hire needed to cover the costs included in the specified breakeven measure.
Brokerage	A commercial fee calculated as a percentage of gross charter revenue.
Browser storage	Data stored by the browser for the application's origin. Clearing this storage removes local studies.
CAPEX	Capital expenditure. CapeAI deducts the entered payment in its specified year.
Carry	The sponsor's stated percentage of positive aggregate nominal equity profit, charged at exit.
Cash deficit	A negative cash amount under the stated basis. Annual investor deficits require additional equity in this model.
Cash flow	A cash inflow or cash outflow at a specified model time.
CFADS	Cash flow available for debt service under CapeAI's stated operating basis. A lender may use another definition.

Term	Meaning in CapeAI
Charter	An agreement for the use of a vessel. The model uses a supplied daily hire assumption.
Charter rate	Hire earned per earning day, in USD/day.
Closing debt	Debt remaining after the year's scheduled principal, maturity repayment, and sale settlement.
Contribution	Cash supplied by equity investors. It is negative in the investor cash-flow stream.
Cost days	Days used to accrue operating costs in each model year.
Debt	Borrowed acquisition funding. CapeAI models one fixed-rate loan.
Debt arrangement fee	A time-zero charge calculated on the initial debt draw.
Debt draw	The loan cash received at acquisition.
Debt service	Interest plus the principal repayment included in the stated service measure.
Declining-value exit	A sale assumption calculated by reducing purchase price at a constant annual percentage.
Depreciation	A reduction of a modeled asset basis. Exit-value decline and tax depreciation are separate assumptions.
Discount factor	The multiplier that converts a future cash flow to time-zero value.
Discount rate	The required annual return for the cash-flow basis being valued.
Disposal cost	A selling cost calculated as a percentage of gross terminal vessel value.
Distribution	Positive investor cash received during the model.
DWT	Deadweight tonnage. CapeAI stores this descriptive vessel field but does not use it in financial calculations.
Downside case	A comparison with less favorable selected assumptions. It is not a probability forecast.
DSCR	Debt-service coverage ratio. It compares operating cash available with the defined debt-service amount.
Earning days	Days that generate charter revenue in a model year.
Equity	Investor funding after debt has funded its specified part of the purchase.
Equity cash flow	Cash paid by or returned to investors after the model's financing, fees, tax, and carry.
Equity discount rate	The required annual investor return used to discount equity cash flows.
Equity IRR	The internal rate of return calculated from the final investor cash-flow stream.
Equity multiple	Total positive investor distributions divided by all investor contributions. It does not measure timing.
Equity NPV	The net present value calculated from final investor cash flows and the equity discount rate.
Exit	Disposal of the vessel at the end of the holding period.

Term	Meaning in CapeAI
Exit proceeds	Vessel sale cash after the specific deductions shown. Gross and net proceeds must remain distinct.
Fee	A charge with a stated percentage or cash basis. A fee is not automatically included in another cost input.
Fixed interest rate	An annual borrowing rate that remains unchanged throughout the loan.
Gross revenue	Charter rate multiplied by earning days, before brokerage and other costs.
Holding period	The number of complete model years from acquisition to vessel disposal.
Hurdle rate	A required return used to judge an investment. CapeAI's simple carry has no hurdle waterfall.
Inflation	The annual percentage change applied to operating costs.
Initial equity	Investor cash required at acquisition after the debt draw.
Interest	A financing expense calculated from opening debt and the annual loan rate.
IRR	Internal rate of return. The annual discount rate that makes the specified cash-flow stream's NPV zero.
JSON	A structured text format used to export and import a study. It is not encryption.
LDT	Light displacement tonnage. It is the physical quantity used for the entered scrap-value calculation.
Legacy study	A study created by an earlier application with another data or calculation structure.
Leverage	The use of debt to fund part of the acquisition. It changes investor cash flows and risk.
Liquidity	The ability to meet cash payments when due. Positive lifetime profit does not establish annual liquidity.
Local draft	The current editable study retained by the browser, when storage is available.
LTV	Loan-to-value percentage. In CapeAI, it means initial debt divided by vessel purchase price.
Management fee	An annual charge calculated on original vessel purchase price in this model.
Manual exit	A terminal vessel value entered directly by the user.
Maturity	The model year when all remaining contractual debt becomes due.
Model version	An identifier for the calculation rules used by a release.
MOIC	Multiple on invested capital. CapeAI uses its defined equity multiple.
Named study	A saved entry in the browser's study library. It is separate from the editable draft.
Net cash flow	Inflows less outflows for the stated period and basis.
Nominal amount	A cash amount stated in the currency units expected for its payment period.
Nominal profit	Undiscounted distributions less contributions. It does not adjust for the time value of money.

Term	Meaning in CapeAI
NPV	Net present value. The discounted sum of all cash flows, including the initial investment.
Off-hire	Time without charter revenue. Lower earning days can represent its effect.
Opening debt	The debt balance before the current year's interest and principal payments.
Operating funding breakeven	The daily charter rate required to cover the defined annual operating and contractual financing cash requirements.
Operating surplus	Gross charter revenue less OPEX, brokerage, and the annual management fee.
OPEX	Operating expenditure, entered as a daily cost and accrued over cost days.
Outstanding debt	The loan principal still owed at a specified time.
Principal	The borrowed amount, excluding interest and financing fees.
Required return	The return an investor or capital provider requires for the assumed risk.
Residual value	The vessel's assumed value at disposal. CapeAI uses the term terminal value for this amount.
Return basis	The cash flows and fees included in a reported return.
ROI	Return on investment. Where used, CapeAI defines it as nominal equity profit divided by total equity contributions.
Scenario	A complete recalculation using specified changed assumptions.
Scheduled principal	The equal annual principal amount due before any additional balloon or early-sale settlement.
Schema version	An identifier for the structure of an exported study.
Scrap price	The assumed USD value per light displacement tonne at disposal.
Scrap value	Light displacement tonnage multiplied by the entered scrap price.
Sensitivity	The change in an output when selected assumptions change.
Sign change	A switch between negative and positive nonzero cash flows. Several switches can make IRR interpretation ambiguous.
SPV	Special purpose vehicle. A legal entity used to hold an investment; CapeAI does not create one.
Study	One named set of vessel-investment assumptions and its calculated results.
Survey	A scheduled inspection or related maintenance event. The analyst supplies its cash cost and payment year.
Tax basis	The modeled amount available for future tax depreciation and sale-gain calculations.
Tax depreciation	A noncash deduction calculated from the opening modeled tax basis.
Tax rate	The user-entered percentage used by CapeAI's simplified tax formula.
Tenor	The loan's contractual duration in complete model years.

Term	Meaning in CapeAI
Terminal value	Gross vessel value at the end of the holding period, before selling costs and debt repayment.
Time zero	The acquisition date in the annual model, before the first operating year.
Unlevered	Calculated before debt funding, debt service, and debt-related tax benefits.
Upside case	A comparison with more favorable selected assumptions. It is not a probability forecast.
Validation	Checks that inputs satisfy the supported types and model limits. It does not verify market plausibility.
Working capital	The entered initial operating funding amount, held constant and released fully at exit.
Year zero	Another name for time zero.

Unit and notation rules

Notation	Meaning
`USD` or `$`	United States dollars.
`USD/day`	Dollars per day.
`USD/tonne`	Dollars per metric tonne.
`%`	Percentage points entered by the user. Enter 7 for 7%.
Decimal fraction	A percentage divided by 100 for use in equations.
`×`	Multiplication.
`Σ`	Sum over the specified periods or items.
`max(a, b)`	The greater of `a` and `b`.
`min(a, b)`	The smaller of `a` and `b`.
`t`	The current year index. Time zero has index `0`.
`H`	The final holding year.
Subscript	A variable's period or basis identifier.

Terms to keep distinct

Price and value. Price is the purchase input. Discounted cash-flow value depends on future cash flows and the required return.

Profit and liquidity. A profitable study can require more cash before the vessel is sold.

Depreciation and CAPEX. Depreciation affects the simplified tax base. CAPEX is an actual modeled cash payment.

Asset return and equity return. Debt changes investor funding and payments. It does not change the vessel's gross charter revenue.

Scenarios and forecast probabilities. The tool recalculates entered assumptions without assigning their likelihood.

REFERENCE BOOK / 03

Original system and forensic findings

Status: historical description, before the rebuild.

Reviewed snapshot: `e45add581ef9adc3a84ff203363e428c3b29225b`.

Review date: 18 October 2026.

Audience: analysts, owners, developers, and reviewers.

This document describes observed behavior in the original application. It does not describe the replacement release.

Read Target system for the intended correction. Read User and model manual for the replacement's actual behavior.

1. Purpose and operating model

The original application screens one vessel acquisition. A user supplies vessel details, financing terms, operating assumptions, and an expected exit value.

The browser calculates annual cash flows and investment metrics. No live market-price service supplies its financial assumptions.

Its intended journey has eleven sections. These cover the vessel, acquisition, costs, surveys, charter income, cash flow, SPV fees, sensitivity, and a conclusion.

Vessel descriptions provide context. Purchase price, financing, income, costs, and exit assumptions drive the numerical results.

The application offers study saving, JSON export, printing, and presentation export. Several features have defects described below.

2. Three calculation paths

Path	Location at the reviewed snapshot	Role
Root application	`index.html`	The observed production website.
Presentation generator	A second script in `index.html`	A separate financial model for PowerPoint output.
React application	`app 2/src/lib/calculations.ts`	An alternative application with separate defaults and calculations.

These paths do not share one financial engine. Their definitions and results differ.

The reviewed production HTML matched the repository file byte for byte. Its SHA-256 was `bbc030786ec90e0f7c2ff7a4825720154d1034f9a21b7f949d1978e14ead8109`.

3. Inputs and their original role

Input group	Main fields	Original use
Vessel	Name, type, DWT, build year, flag, class, specifications, LDT	Identification; LDT feeds scrap value.
Acquisition	Price, equity percentage, debt percentage, ownership vehicle	Initial funding and investment structure.
Debt	Interest rate, tenor, balloon percentage	Annual interest and principal schedule.
Operations	Crew, stores, repairs, insurance, management, miscellaneous costs	Annual operating expense estimate.
Cost escalation	OPEX inflation and optional daily override	Future operating expenses.
Surveys	Special survey, intermediate survey, other CAPEX	A pooled budget divided across the holding period.
Charter	Daily hire, earning days, annual overrides	Base revenue; overrides affect a separate table only.
Timing	Holding period, OPEX days	Number of projected years and cost accrual.
Exit	Depreciation mode, scrap mode, manual sale value	Terminal vessel value, except in the presentation model.
SPV	Management fee, commercial fee, carry, minimum investment	Annual fees and nominal-profit sharing.
Commentary	Risks, mitigations, analyst, recommendation	Presentation text; no independent underwriting assessment.

The engine omits explicit transaction costs, working capital, tax, and NPV. It has no contractual refinancing schedule.

The survey-date fields do not establish dated payments. The calculation spreads the pooled survey budget across every holding year.

4. Original financial mechanics

Use the following symbols to read this section:

Symbol	Meaning	Unit
P	Purchase price	USD
d, b, i	Debt fraction, balloon fraction, annual interest rate	Decimal fraction
T, H	Loan tenor and holding period	Years
h, e	Base charter hire and annual earning days	USD/day; days/year
D, B, A	Initial debt, original balloon, scheduled annual principal	USD

The debt schedule starts with these equations:

```
D = P × d

B = D × b

A = (D - B) / T

Annual interest = opening debt balance × i
```

The principal model calculates base revenue as `h × e` in every year. Annual charter overrides do not feed these cash flows.

The initial cash outflow includes equity only. Later cash flows deduct debt service.

Therefore, the original label “Project IRR” describes a leveraged equity return. It does not describe an unlevered asset return.

At exit, the model subtracts the original balloon from vessel sale value. It does not settle the actual outstanding debt correctly for all timings.

The original carry deducts a percentage of positive aggregate nominal profit. It has no preferred-return hurdle or detailed investor waterfall.

The original financial source records these mechanics.

5. Confirmed defects and their effect

Defect	Reproducible effect	Required correction
Annual charter overrides bypass the return model	Lowering one year's hire changes its charter table but leaves headline returns unchanged.	Use annual revenue inputs in every output.
Early sale deducts only the balloon	A five-year sale can omit remaining scheduled principal on a ten-year loan.	Settle the actual outstanding loan.
Late maturity handling is incomplete	Holding beyond maturity does not fund the balloon at its due date.	Pay all remaining debt at maturity.
Breakeven uses the wrong day basis	A displayed breakeven hire rate can still produce a cash loss.	Divide required annual revenue by earning days.
Zero inputs become defaults	Zero debt, inflation, fees, or hire can become positive assumptions.	Preserve explicit zero values.
Metrics use different fee bases	Headline IRR, charts, and sensitivity can describe different cash-flow streams.	Label each basis and share one engine.
IRR lacks failure handling	Cash flows `[-100, 0]` return 10%.	Return an unavailable state when no valid solution exists.
Export has a separate engine	Manual and scrap exits can differ between the screen and presentation.	Generate reports from the same calculated result.
PowerPoint script redeclares a constant	The script fails before its export function initializes.	Retire or replace the faulty exporter.

Example: overstated early-sale proceeds

Assume a $10 million purchase and 50% debt. The loan has a ten-year tenor and a 30% balloon.

Initial debt is $5 million. The original balloon is $1.5 million. Annual principal is $350,000.

After five annual principal payments, debt remains $3.25 million. The original model deducts $1.5 million at sale.

The resulting equity sale proceeds are overstated by $1.75 million. This difference directly distorts investment returns.

Example: false breakeven

Assume daily OPEX of $6,000, 365 cost days, and 350 earning days. Set debt, surveys, and fees to zero.

Annual OPEX is $2,190,000. The original displayed breakeven is $6,000/day.

At that hire rate, annual revenue is $2,100,000. The vessel loses $90,000 before fees.

The cost-covering rate is `$2,190,000 / 350 = $6,257.14/day`.

Example: disconnected charter forecast

Use a $15,000/day base hire and 350 earning days. Reduce year-one hire to $5,000/day.

The charter table loses $3.5 million of year-one revenue. The original headline return calculation does not change.

These examples use invented inputs. They do not describe a real vessel investment.

6. Storage, privacy, and unfinished functions

The root application includes Supabase sign-in and cloud study saving. Its study serializer collects every input, select, and text area.

This includes the hidden email and password fields. Offline execution confirmed that dummy credentials entered the export and cloud-save payloads.

The review did not inspect historical cloud records. It did not establish whether real passwords were stored.

Study names and other external values enter unsafe HTML rendering paths. Backend row-level access policies are absent from the repository.

The public Supabase anonymous key is not an administrator secret. Its presence does not prove a database-access defect.

The root AI functions lack a complete authenticated integration. The React alternative uses demonstration extraction and a templated conclusion.

The reviewed application provides no verified market data, document intelligence, or investment advice service.

See the original export serializer and cloud serializer.

7. How to use historical studies during migration

1. Keep a separate copy of the original study file.
2. Import a copy into the replacement application.
3. Read each migration warning.
4. Check every financing and exit assumption.
5. Assign survey payments to their actual years.
6. Review the annual charter forecast.
7. Recalculate the study with the replacement engine.
8. Export a new versioned study file.

Do not treat matching old and new IRRs as a migration requirement. Corrected debt, fee, and cash-flow timing can change the result materially.

8. Review boundary

The forensic work combined source inspection, isolated function tests, and read-only production checks. It did not certify the original software.

No live study, account, database record, or private financial document was changed during that review.

The original repository history remains the evidence source. The replacement manual records the new implementation separately.

REFERENCE BOOK / 04

Target system and acceptance criteria

Status: design specification for the replacement release.

Decision date: 18 October 2026.

Product model: device-first calculator, without accounts.

This document defines the intended behavior. The release manual describes what the implementation supports.

1. The job to complete

An analyst must understand the cash consequences of buying, operating, financing, and selling one vessel.

The tool must expose assumptions and funding needs. Every headline metric must reconcile to a visible annual cash-flow stream.

The tool must help users compare cases. It must not convert uncertain forecasts into an automatic investment recommendation.

2. User journey

Stage	User action	Required result
Start	Open the example or create a blank study.	An editable study with clear units and validation.
Study setup	Enter acquisition, income, operating, financing, tax, and exit assumptions.	One consistent set of model inputs.
Cash flow	Inspect annual operations, debt, sale proceeds, and investor cash.	A reconciled explanation of returns and funding gaps.
Scenarios	Compare charter-rate cases; edit exit assumptions in a separate study.	Comparable outputs from the same engine.
Manual & equations	Read equations, definitions, and limits.	An explanation matching the implemented model.
Device library and Export	Save locally or export a study and report.	Portable data and a reproducible record.

The desktop interface must support detailed comparison. The mobile interface must preserve the same calculations and essential actions.

Large financial tables may scroll horizontally. The page layout must remain usable without horizontal page overflow.

3. Financial contract

The detailed implementation contract is Model contract.

Area	Required behavior
Currency	Use USD consistently; do not imply an exchange-rate service.
Timing	Use time zero plus annual, end-of-year cash flows.
Acquisition	Include purchase price, transaction costs, initial working capital, and initial CAPEX.
Debt funding	Calculate debt from purchase price; fund the remaining acquisition uses with equity.
Debt fees	Charge a stated percentage of initial debt at time zero.
Revenue	Use each year's actual rate and earning days.
OPEX	Apply the cost-day basis and explicit annual inflation.
CAPEX	Pay each scheduled amount in its selected year.
Annual fees	Define brokerage and management fee bases separately.
Debt service	Charge interest on opening debt; pay scheduled principal at year end.
Maturity	Settle all remaining debt at maturity without assumed refinancing.
Early exit	Settle the actual remaining debt after that year's scheduled principal.
Exit	Support explicit manual, declining-value, and scrap assumptions.
Working capital	Fund the initial amount and release that amount at exit.
Tax	Provide an explicit simplified assumption; document jurisdictional exclusions.
Carry	Share positive cumulative nominal equity profit through an explicit final-year charge.
Asset return	Calculate cash flow before debt financing and investor carry.
Equity return	Calculate cash flow after debt, fees, tax assumptions, and carry.
NPV	Match the asset rate to asset cash flows and equity rate to equity cash flows.
IRR	Report unavailable or ambiguous results when appropriate.
Liquidity	Show extra equity contributions when annual cash is negative.
Breakeven	Use earning days and disclose excluded terminal items.

Zero is a valid assumption where the input permits it. Missing values must not silently replace explicit zero values.

4. Data and privacy contract

The first release has no account, cloud study database, external AI service, or market-data integration.

Study content stays in the browser until the user exports or shares it. Browser storage is not an encrypted vault.

Study JSON must have a version. Imports must validate types, bounds, and supported fields before use.

Imported content must never execute as HTML or script. Credential fields must never become study fields.

Legacy imports must report changed assumptions. A legacy file must not silently inherit obsolete calculation behavior.

Local saving does not replace backups. A downloaded study file must support recovery after browser storage is cleared.

5. Report contract

The screen, sensitivity analysis, CSV, printable report, and PowerPoint deck must use the same financial engine.

Reports must identify the study, model version, assumptions, cash-flow basis, and calculation limitations.

The release replaces the original presentation engine with a local PowerPoint exporter. Its results must match the screen's financial engine.

6. Design contract

Controls must show units beside values. Error messages must identify the field and explain the necessary correction.

Navigation must distinguish input preparation from result inspection. The current stage must remain visible.

The interface must support keyboard operation, visible focus, readable contrast, and comfortable touch targets.

Long explanations belong in field help and the manual. Essential basis labels belong beside the corresponding financial result.

7. Verification gates

Gate	Evidence required
Financial correctness	Independent examples and regression tests for every material forensic defect.
Reconciliation	Initial funding balances, debt closes at zero, and cash-flow identities hold.
Zero assumptions	All-equity, zero-fee, zero-inflation, and no-hire cases remain explicit.
Debt timing	Early sale and maturity before sale both settle the correct debt.
Return limits	No-recovery and multiple-sign-change cash flows do not display invented IRRs.
Import safety	Invalid fields fail validation; known legacy data maps deliberately.
Browser journey	Create, edit, inspect, save, reload, import, export, and print work.
Responsive layout	Core actions work on desktop and mobile viewports.
Documentation	Equations and examples match the released engine.
Deployment	The production URL loads and its core journey works.

8. Deliberate scope limits

The release is a deterministic annual screening model. It is not a monthly treasury model, accounting ledger, or fleet optimizer.

It does not price vessels or forecast charter markets. It does not provide tax advice or a contractual loan-covenant opinion.

The release does not model refinancing, floating rates, derivative hedges, exchange rates, probabilistic simulations, or a preferred-return waterfall.

Future additions require clear definitions and independent test cases. Their absence must remain visible in the manual.

REFERENCE BOOK / 05

Financial model contract

Engine version: **1.0.0**. Study schema version: **1**.

`src/model.mjs` is the only calculation engine. Interfaces, scenarios and reports must consume its results. They must not reimplement formulas.

Conventions

- All currency amounts are USD. The model performs no currency conversion.
- Input fields ending in `Pct` contain whole percentages. For example, `55` means 55%.
- Output IRR and ROI values are decimal fractions. For example, `0.10` means 10%.
- `calculateNPV` accepts a decimal discount rate.
- Time zero records acquisition, financing and initial funding.
- Years 1 through `holdYears` represent equal annual periods.
- All annual operating payments, taxes and distributions occur at year end.
- Exit occurs after the final year's operations, tax depreciation and contractual principal payment.
- Positive investor cash flow means a distribution. Negative investor cash flow means an equity contribution.
- Annual inflows and outflows are netted. The model does not estimate funding needed within a year.
- Amounts remain unrounded in the engine. Round only for presentation.
- Missing fields are validation errors. Explicit zero values remain zero.

Public API

```
createBlankStudy() // Complete schema; price 0 requires user input.
createExampleStudy() // Complete, valid illustrative assumptions.
validateStudy(study) // [{ path, severity: 'error'|'warning', message }]
calculateStudy(study) // Model described below.
calculateNPV(rate, cashflows) // number|null; cashflows start at time zero.
calculateIRR(cashflows) // { value: number|null, status, message }
calculateSensitivity(study, multipliers = [0.7, 0.85, 1, 1.15, 1.3])
```

`calculateStudy` does not mutate its input. A valid result contains a copied study.

An invalid result has `valid: false`, validation issues, empty rows and flows, and null summaries. It does not echo invalid study data.

Unknown fields are rejected, including authentication fields. Import adapters must explicitly map legacy data into the versioned schema before calculation.

Study fields

Every field below is required. Empty arrays and permitted zero values are valid.

Path	Meaning and bounds
`schemaVersion`	Exactly 1
`name`	Study label, at most 200 characters
`notes`	User notes, at most 10,000 characters
`vessel.name`, `vessel.type`	Text, at most 200 characters each
`vessel.builtYear`	Integer 0-2200; zero means unspecified; informational
`vessel.deadweight`	Tonnes, 0-1,000,000; informational
`vessel.ldt`	Light displacement tonnes, 0-1,000,000; used in scrap value
`acquisition.price`	Vessel purchase price, $0.01-$1 trillion
`acquisition.transactionCosts`	Fixed acquisition costs, $0-$1 trillion
`acquisition.workingCapital`	Initial operating cash investment, $0-$1 trillion
`financing.ltvPct`	Debt as a percentage of vessel price, 0-100%
`financing.interestPct`	Fixed annual interest on opening debt, 0-100%
`financing.tenorYears`	Contractual loan maturity, integer 1-30 years
`financing.balloonPct`	Percentage of original debt due after scheduled amortization, 0-100%
`financing.arrangementFeePct`	Time-zero fee as a percentage of debt drawn, 0-100%
`operations.dailyOpex`	Base daily vessel operating costs, $0-$100 million
`operations.opexDays`	Annual cost days, integer 0-365
`operations.inflationPct`	Compound annual change in OPEX, -100-100%
`operations.baseDailyRate`	Base daily charter rate, $0-$100 million
`operations.baseRevenueDays`	Base earning days, integer 0-365
`operations.brokeragePct`	Annual brokerage as a percentage of gross revenue, 0-100%
`operations.managementFeePct`	Annual SPV fee as a percentage of original vessel price, 0-100%
`projection.holdYears`	Integer 1-30 years
`projection.assetDiscountPct`	Discount rate for asset cash flows, 0-100%
`projection.equityDiscountPct`	Discount rate for each equity cash-flow basis, 0-100%
`charter`	At most 30 `{ year, dailyRate, revenueDays }` rows
`charter[].year`	Unique integer 1 through holding period
`charter[].dailyRate`	Annual override, $0-$100 million
`charter[].revenueDays`	Annual override, integer 0-365
`capex`	At most 300 `{ year, amount, label }` rows; repeated years are allowed

Path	Meaning and bounds
`capex[].year`	Payment year, integer 0 through holding period
`capex[].amount`	Nominal payment, $0-$1 trillion
`capex[].label`	Text, at most 200 characters
`exit.mode`	`depreciation`, `manual` or `scrap`
`exit.depreciationPct`	Annual decline for the depreciation exit estimate, 0-100%
`exit.manualValue`	User's gross sale estimate, $0-$1 trillion
`exit.scrapPerTon`	Scrap price per light displacement tonne, $0-$100 million
`exit.disposalCostPct`	Sale cost as a percentage of gross exit value, 0-100%
`tax.ratePct`	Simplified tax rate, 0-100%
`tax.annualDepreciationPct`	Annual declining tax-book depreciation rate, 0-100%
`tax.taxSaleGain`	Boolean: include positive disposal gains in taxable income
`carry.ratePct`	Whole-deal nominal profit share, 0-100%

All numeric values must be finite numbers, not numeric strings. Unknown object keys are errors.

Text rejects unsupported control characters, unpaired UTF-16 surrogates, and U+FFFE/U+FFFF. Tabs, line breaks, and complete Unicode characters are permitted.

When holding period changes, charter and CAPEX years outside the new period require explicit correction. The engine does not discard them.

Warnings do not block calculation. Current input warnings cover enabled simplified tax and zero light displacement for scrap valuation.

Funding and debt

Use these symbols:

- `P`: acquisition price.
- `A`: acquisition transaction costs.
- `W`: initial working capital.
- `C0`: sum of CAPEX payments at time zero.
- `D`: initial loan draw.
- `F`: loan arrangement fee.
- `H`: holding period.
- `T`: loan tenor.
- Percentage symbols below denote decimal fractions.

```
D = P × LTV
F = D × arrangement-fee rate
Asset uses = P + A + W + C0
Total initial uses = Asset uses + F
Initial equity = Total initial uses - D
Annual scheduled principal = D × (1 - balloon rate) / T
```

Debt finances the vessel price only. Acquisition costs, initial CAPEX, working capital and arrangement fees require equity.

Each year follows this order:

1. Calculate interest on the opening loan balance.
2. Pay scheduled principal, limited to the remaining balance.
3. At maturity, repay every remaining dollar as `maturityPrincipal`.
4. At exit, settle any balance still outstanding as `exitDebtSettlement`.

No refinancing is assumed. A balloon paid at maturity can require additional equity before disposal.

On early sale, settlement includes all outstanding scheduled principal and balloon. On sale at maturity, no duplicate settlement occurs.

```
Interest = opening balance × interest rate
Total principal = scheduled principal + maturity principal + exit settlement
Debt service = interest + total principal
Closing balance = opening balance - total principal
```

Operations and fees

An override row replaces both the base daily rate and earning days for its year. Without a row, both base assumptions apply.

Explicit zero days or a zero rate produce zero revenue. They never restore the base assumptions.

```
Revenue[y] = daily charter rate[y] × earning days[y]
OPEX[y] = daily OPEX × annual cost days × (1 + inflation)^(y-1)
Brokerage[y] = revenue[y] × brokerage rate
Management[y] = original vessel price × management-fee rate
Operating surplus[y] = revenue[y] - OPEX[y] - brokerage[y] - management[y]
CAPEX[y] = sum of nominal payments assigned to year y
```

Management fees do not inflate or decline with vessel value. Brokerage changes with each year's actual revenue.

CAPEX is paid in its selected year. It is not amortized into an invented annual cash payment.

Exit and working capital

```
Depreciation-mode gross value = P × (1 - exit depreciation)^H

Manual-mode gross value = manual value

Scrap-mode gross value = light displacement tonnes × scrap price per tonne

Disposal costs = gross value × disposal-cost rate

Net sale proceeds = gross value - disposal costs
```

These are alternative user assumptions. They do not constitute an independent valuation or use live market data.

CAPEX does not automatically increase the gross sale estimate. Use a manual exit estimate when that effect matters.

All initial working capital returns at exit at its original nominal amount. No interest, growth or impairment is modeled.

`exit.debtSettlement` reports only the additional early-sale balance settlement. Final-year contractual principal remains in the annual debt schedule.

Simplified tax

Tax is zero unless the user enables a positive rate. This is an illustrative calculation, not a jurisdiction-specific tax model.

```
Initial tax book = P + acquisition transaction costs + year-zero CAPEX

Tax depreciation[y] = opening tax book[y] × annual tax-depreciation rate

Closing tax book[y] = opening book - depreciation + current-year CAPEX

Taxable disposal gain = max(0, net sale proceeds - closing tax book)
```

Current-year CAPEX joins the book at year end. It starts depreciation in the following year.

The disposal gain is included only in the exit year when `taxSaleGain` is true. Disposal losses are not deducted.

Let `G` be that year's taxable disposal gain, or zero otherwise.

```
Asset tax = tax rate × max(0, operating surplus - tax depreciation + G)

Equity tax = tax rate × max(0, operating surplus - interest - tax depreciation + G)

Pre-fee equity tax = tax rate × max(0, revenue - OPEX - interest - tax depreciation + G)
```

Operating losses can offset a positive disposal gain within the same year. No loss carries forward or back; there are no tax refunds.

Principal, working capital and carry are not tax deductions. Arrangement fees are not deducted or amortized for tax.

Tax is recalculated for each cash-flow basis. Removing a deductible annual fee can therefore increase tax in the pre-fee comparison.

Cash-flow bases and carry

Each cash-flow array begins with its time-zero outlay.

Basis	Time zero	Annual flow
`asset`	Negative asset uses	Operating surplus - CAPEX - asset tax + terminal cash
`preFees`	Negative initial equity	Revenue - OPEX - CAPEX - pre-fee equity tax - debt service + terminal cash
`preCarry`	Negative initial equity	Operating surplus - CAPEX - equity tax - debt service + terminal cash
`investor`	Negative initial equity	Pre-carry equity cash flow - carry payment

Terminal cash is net sale proceeds plus working-capital release. It is zero before exit.

Asset flows include brokerage and SPV management as operating expenses. They exclude debt proceeds, debt payments, arrangement fees and carry.

The pre-fee equity comparison excludes annual brokerage, management and carry. It retains loan arrangement costs in the initial equity outlay.

```
Whole-deal nominal profit = sum of every pre-carry equity flow, including time zero
Carry = max(0, whole-deal nominal profit) × carry rate
Final investor flow = final pre-carry flow - carry
```

Negative annual flows reduce the carry base. No preferred return, hurdle, catch-up or interim carry is modeled.

Carry is paid at exit, even if that requires a final equity contribution. Earlier distributions are not automatically held in reserve.

Return metrics

Every basis has the same metric shape:

```
invested = absolute sum of negative cash flows
distributions = sum of positive cash flows
profit = distributions - invested
multiple = distributions / invested
ROI = profit / invested
NPV(r) = sum from t=0 through H of cashflow[t] / (1+r)^t
```

Multiple and ROI are null when invested capital is zero. Additional contributions count as invested capital.

Asset NPV uses `assetDiscountPct`. All three equity NPVs use `equityDiscountPct`. The engine does not estimate either required return.

IRR is the annual rate that sets NPV to zero. It needs an initial outflow pattern followed by positive receipts.

The solver ignores exact zero entries when counting sign changes. With one change from negative to positive, a unique investment IRR exists.

With more than one sign change, the engine reports `ambiguous`. This conservative label means uniqueness is not assured; it does not assert multiple real roots.

Without both positive and negative flows, IRR is `unavailable`. Unrepresentable or unreliable numerical roots are also unavailable.

The solver uses bracketed bisection in `log(1+r)` and checks the residual. It never returns an unconverged starting guess.

DSCR and operating funding breakeven

These measures exclude sale proceeds and working-capital release. They also exclude carry and additional early-sale debt settlement.

Operating tax excludes disposal gains:

```
Operating tax = tax rate × max(0, operating surplus - interest - tax depreciation)

Operating cash flow = operating surplus - actual CAPEX - operating tax

Contractual debt service = interest + scheduled principal + maturity principal

DSCR = operating cash flow / contractual debt service
```

DSCR is null when contractual debt service is zero. At maturity, its denominator includes the balloon.

Operating funding breakeven is the minimum daily charter rate that covers the year's cash requirements without sale proceeds.

Let:

```
K = CAPEX + interest + scheduled principal + maturity principal

S = interest + tax depreciation

q = tax rate

b = brokerage rate

d = actual earning days
```

Required operating surplus `X` follows:

```
If K ≤ S: X = K

If K > S and q < 1: X = (K - qS) / (1 - q)

Required daily rate = (OPEX + management + X) / (d × (1 - b))
```

The result is null for zero earning days or 100% brokerage. It is also null when 100% tax prevents a finite solution.

This metric uses actual earning days. It can rise sharply in a CAPEX or loan-maturity year.

Result shape

```
{
  valid, issues, engineVersion, study,
  rows, // Years 1..hold; includes all components below.
  flows: { asset, preFees, preCarry, investor }, // Arrays include time zero.
  metrics: { asset, preFees, preCarry, investor },
  funding: { initialUses, initialDebt, initialEquity, arrangementFee, upfrontCapex },
  exit: { grossValue, disposalCosts, netValue, debtSettlement,
          workingCapitalRelease, taxBookValue, taxableGain },
  totals: { revenue, opex, capex, brokerage, management, interest, principal,
            tax, carry, additionalEquity, distributions },
  warnings // Plain text; input and calculated conditions.
}
```

Each `metrics` item contains `irr`, `npv`, `discountPct`, `invested`, `distributions`, `profit`, `multiple` and `roi`.

Each `irr` contains `value`, `status` (`ok`, `unavailable`, or `ambiguous`) and `message`.

Each annual row contains:

```
year, dailyRate, revenueDays, revenue, opex, brokerage, management, capex,
taxDepreciation, taxBookValue, assetTax, equityTax, preFeeTax,
interest, scheduledPrincipal, maturityPrincipal, exitDebtSettlement,
principal, debtService, debtOpening, debtClosing,
saleValue, disposalCosts, saleNet, workingCapitalRelease,
assetCashFlow, equityPreFees, equityPreCarry, carry, investorCashFlow,
contribution, distribution, operatingCashFlow, dscr, breakevenDailyRate,
cumulativeInvestorCashFlow
```

`contribution = max(0, -investorCashFlow)`. `distribution = max(0, investorCashFlow)`.

Cumulative investor cash flow includes the initial equity outlay. It is not cumulative gross distributions.

Sensitivity

Sensitivity multiplies every base and overridden daily charter rate by the selected factor. It preserves earning days and all other assumptions.

Each point runs `calculateStudy`. It reports the after-carry investor IRR, NPV, multiple and additional equity.

Scenario results therefore use the same fees, tax, debt, CAPEX and carry rules as the main report.

Regression evidence

Run `node --test tests/model.test.mjs`.

Independent reference cases cover zero inputs, charter overrides, no-hire years, early sale, maturity, terminal modes, fees, tax and carry.

They also cover timed CAPEX, funding reconciliation, additional contributions, consistent investor metrics and invalid or ambiguous IRR.

The tests verify investor cash-flow identities and total debt repayment. Model output must remain finite or explicitly null.

Scope limits

The model excludes monthly payment dates, day-count conventions, floating rates, refinancing and covenant tests.

It excludes detailed jurisdictional taxes, tax-loss carryforwards, currency conversion, hedging, reserve-account interest and intra-year liquidity.

It does not forecast charter markets, vessel values, technical condition, survey dates or downtime. Users must enter those assumptions.

Negative equity flows assume the investor can fund every shortfall. The model does not enforce limited liability or simulate insolvency.

There is no automated investment recommendation. Positive NPV depends on the user's assumptions and chosen required return.

Primary references

The cash-flow and discount-rate separation follows the principles in Damodaran's Approaches to Valuation.

Cash-flow timing and required returns are also discussed in First Principles.

The annual timing, tax approximation and carry convention above are CapeAI implementation choices. They are not presented as Damodaran's prescribed vessel model.

REFERENCE BOOK / 06

References and source boundaries

Reviewed on 18 October 2026.

These sources establish general principles and writing guidance. CapeAI's equations and examples describe its own implementation.

No author or institution listed here has approved, audited, or endorsed CapeAI.

Valuation principles

1. Aswath Damodaran, Approaches to Valuation, NYU Stern. This supports the distinction between asset and equity cash flows. It also supports matching cash-flow and discount-rate bases.
2. Aswath Damodaran, First Principles, NYU Stern. This supports evaluating both the amount and timing of investment cash flows.
3. Aswath Damodaran, Capital Budgeting under Certainty, NYU Stern. This discusses NPV, IRR, multiple IRRs, investment scale, working capital, and debt financing.
4. Aswath Damodaran, Financial Measures and Ratios, NYU Stern. This supports the cash-flow distinction between operating earnings, reinvestment, and financing.
5. Aswath Damodaran, Cash Flows, NYU Stern. This supports treating capital expenditure and working-capital investment as cash requirements.

CapeAI does not use a continuing-business perpetuity for terminal value. Its terminal value is an explicit vessel disposal assumption.

Its simplified tax model is a software assumption. The references do not establish an applicable shipping tax regime.

Writing guidance

6. ASD Simplified Technical English Maintenance Group, ASD-STE100 Issue 9, dated 15 January 2025. The official search index confirmed Issue 9 and its procedural-writing rules. Direct PDF retrieval returned HTTP 403 during this review.
7. ASD Simplified Technical English Maintenance Group, official standard website. This identifies the standard and its maintenance organization.

The project uses an STE-inspired editorial target. At least 80% of measured prose sentences should contain no more than 25 words.

Procedural steps should contain one action and no more than 20 words. Essential financial technical terms remain in the controlled project glossary.

This is not an ASD-STE100 conformity claim. A sentence-length check does not verify the complete grammar rules or approved-word dictionary.

Historical implementation evidence

8. Original repository snapshot. The current-system document describes this fixed snapshot.
9. Original main financial engine.
10. Original presentation financial engine.
11. Original study export serializer.

The forensic findings concern the reviewed snapshot. They must not be presented as defects in a later release without new evidence.